synkit

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A toolkit to create syntactic representations in Typst

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Abstract

synkit is a Typst package for drawing syntax trees from bracket notation. It supports a wide range of features, including triangles, arrows, movement traces, multidominance, semantic annotation, cross-tree equivalence lines, and formatting options for quick adjustments. Trees can be drawn in four directions (down, up, left, right), and multiple trees can be stacked with shared arrow connections. Besides trees, the package also offers functions for numbered examples and glosses. This document serves as the official documentation for **synkit**, providing examples and explanations for all available parameters and features.

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Questions, suggestions, bugs

Any questions, comments or suggestions should be posted to the repository below (issues). All the bugs you find will help improve the package! Simply open an issue.

- `gdgarcia.ca/synkit`
- `guilhermegarcia/synkit`
- `guilhermegarcia/synkit/discussions`
- `guilhermegarcia/synkit/issues`

Note on development

syn*kit* was conceived, designed, and directed by the author. The implementation was produced with the assistance of Claude (Anthropic). All linguistic decisions, function design, and package architecture are the author's.

Version history

`0.0.2` - Bug fixes, literal brackets now accepted as escaped characters

`0.0.1` - Initial release

Table of contents

1. Introduction

This vignette introduces **syn*kit*** and its functions. If you haven't used Typst yet and want to follow along, go to Typst.app to use their online editor. You may want to check their own tutorial too (I have an introductory YouTube series here). This is **not** an introduction to Typst, but see Section B and Section C in the appendix for some useful info. The GitHub repository for the package can be found at `guilhermegarcia/synkit`. Comments and suggestions are welcome, as are bug reports (open an issue in the package's repository). As with **phono*kit***, the main goals of **syn*kit*** are to MINIMIZE EFFORT and MAXIMIZE QUALITY. In a nutshell, we want more intuitive and parsimonious code than LaTeX without compromising the quality of what is created.

There are at least two fantastic packages for LaTeX to build syntax trees: `tikz-qtree` (Chiang, 2009) and `forest` (Živanović, 2017). For Typst, some options also exist: `syntree`, by Lynn (see here) and `arborly`, by Max Pearce Basman (see here). Both Typst options are indeed easier to use than anything we have for LaTeX, but they're also more limited. Thus, as of March 2026, we don't really have a syntax package that ticks enough boxes for people to consider migrating to Typst from LaTeX. This is the gap that **syn*kit*** addresses. My main goal here is to be as intuitive as possible while keeping the key features of packages used in LaTeX. Eventually, **syn*kit*** should be able to cover everything that you need in syntax (which will still be a subset of what the LaTeX packages in question can do, simply because the vast majority of people don't use 100% of what `tikz` can offer in theory, for example). I have some opinions about how an intuitive function should work, so a lot of what follows is subjective — this is also something I've applied to my other package, **phono*kit***. I realize that linguists can't simply migrate to Typst without enough coverage, and that coverage must include things like syntax trees, of course.

I am not a syntactician myself. Thus, if this package ends up being adopted by enough linguists, it would be ideal to have more people in syntax to join the project to maintain the existing functions and features.

1.1. Installation

Typst packages are always loaded the same way: using the `#import` function at the top of your `typ` document, as shown in Code 1. Replace `X.X.X` with the version you wish to import. The `*` simply states that we want to import all functions from the package in question. See the package's page on Typst's website here.

```
#import "@preview/synkit:X.X.X": *
```

Code 1: Loading a package in Typst using the official repository

Alternatively, if you want the most up-to-date version and this version is ahead of the published version, download/clone the repository here and load the package locally. This is shown in Code 2. You simply need to import the `lib.typ` file, which is present in any package.

```
#import "synkit/lib.typ": *
```

Code 2: Loading a package in Typst using a local package

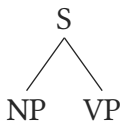
You may need to create a symlink depending on how you structure your files. See Section C for more information on Typst packages in general, as they work slightly differently from what you may be used to if you use LaTeX.

2. Basic trees

Let's begin with a simple tree, following the steps of David Chiang's excellent tutorial on `tikz-qtree` (Chiang, 2009), which you can find here. The function we'll be using is `#tree()`, and by the end of this manual, I hope that all of its parameters will be familiar. The main parameter of the function is the tree itself, which is a string, as can be seen in Code 3. The first thing to notice is that the spaces between `[]` are not a deal breaker.

```
#tree("[S [NP] [VP]]")
#tree("[S[NP][VP]]")
#tree("[ S [ NP ] [ VP ] ]")
```

Code 3: Code for Tree 1 (spaces are very flexible)

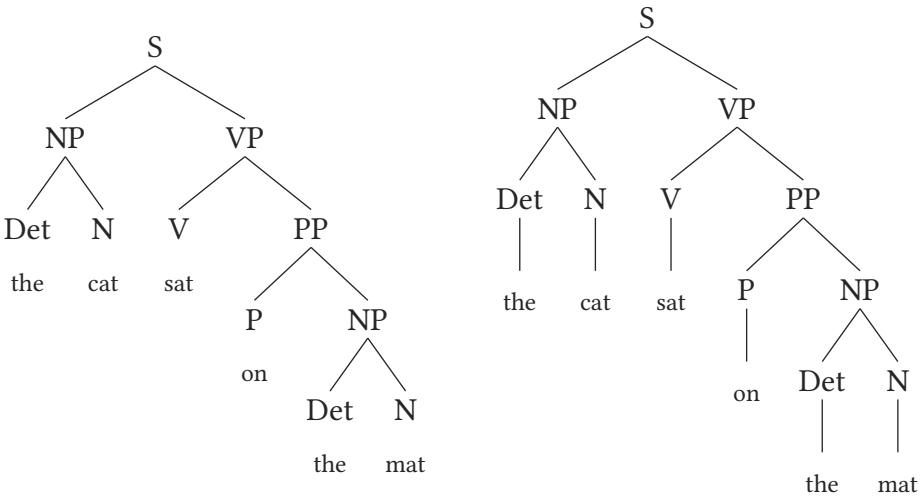


Tree 1: The basics

Next, let's complete the tree following the example in Chiang's tutorial. In Tree 2 (left), no terminal branch is drawn. This is the default behavior of `#tree()`. If you wish to produce these branches, simply add `terminal-branch: true` to the function.

```
#tree("[S [NP [Det the] [N cat]] [VP[V sat][PP[P on] [NP [Det the] [N mat]]]]]")
```

Code 4: Code for Tree 2

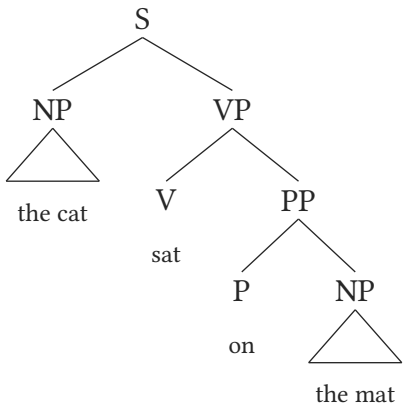


Tree 2: The basics: terminal branches deactivated by default (left)

What about adding a roof for a given XP? You can let the function decide for you whether a triangle is needed *or* you can add it yourself via the parameter `triangle`, which we'll discuss shortly. Examine Code 5: the sequence `[NP the cat]` automatically triggers a triangle because we go from phrase to content without intermediate nodes. This works because the function uses regular expressions to search for a given pattern. Time will tell how reliable this is, but thus far it's been working as it should.

```
#tree("[S [NP the cat] [VP[V sat][PP[P on] [NP the mat]]]]")
```

Code 5: Code for Tree 3

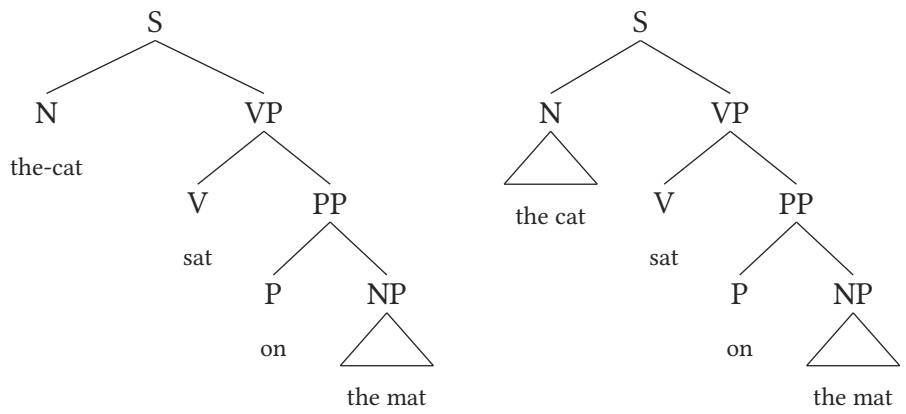


Tree 3: The basics: automatic triangles

How can we manually add triangles? If the function is successful, you will likely never need to do this. But the option is there, and it will help us understand a key element in **synkit**, namely, *labels*. To demonstrate how this works, let's pretend we want to have a triangle under the head `N` (for whatever reason), which is not a context that triggers the automatic drawing just seen. The parameter `triangle` allows you to tell the function which location in the tree should be designed with a triangle. The key here is that labels exist the moment you create a node in the tree: you never have to set labels yourself. In Tree 4, for example, we want a triangle under `N`. This is the first `N` from the top that we see in the tree (well, the only one here). Thus, we must simply state `"N1"` and the `triangle` parameter will target the correct location. This is shown in Code 6. As already mentioned, you should never need to use the `triangle` parameter, but it's there anyway. The notion of automatic labelling is essential for when we discuss arrows.

```
#tree("[S [N the cat] [VP[V sat][PP[P on] [NP the mat]]]]", triangle: ("N1",))
```

Code 6: Code for Tree 4



Tree 4: The basics: custom triangles (`N`)

Terminal node behavior: A hyphen is used in `the-cat` in Tree 4 (left) to ensure that a single atomic string appears under the terminal node. This is recommended for any multi-word expression that should be treated as one terminal, such as `couch-potato` or `pé-de-moleque` (a Brazilian candy, lit. 'boy's foot'). If you instead type `the cat`, a large space will be added between the two words, as if they were sister nodes but without branches. This is by design: `#tree()` assumes that strings in terminal position are atomic. A preterminal category can still take branching structure, which is useful for morphological or compound analyses. For example, `[N [N abc] [N def]]` renders as a noun-noun compound with internal structure, rather than as a single atomic terminal.

3. Arrows

The example below comes from Carnie (2013, p. 261). Arrows are added with the `arrows` parameter, which allows the user to specify `from` and `to` for any given arrow (this uses the automatically created labels already discussed). On top of that, the user can choose `color`, `line-width`, `dash`, and decide whether or not to curve the arrow. By default, arrows are rectangular (see gray dotted arrow in Tree 5) — multiple rectangular arrows have their heights adjusted automatically to avoid overlapping lines. This can be changed with the parameter `curved: true` (outside `arrows`). But if the user provides `bend` and `shift` inside `arrows`, `curved` will be assumed to be `true`. This is what we see in Tree 5.

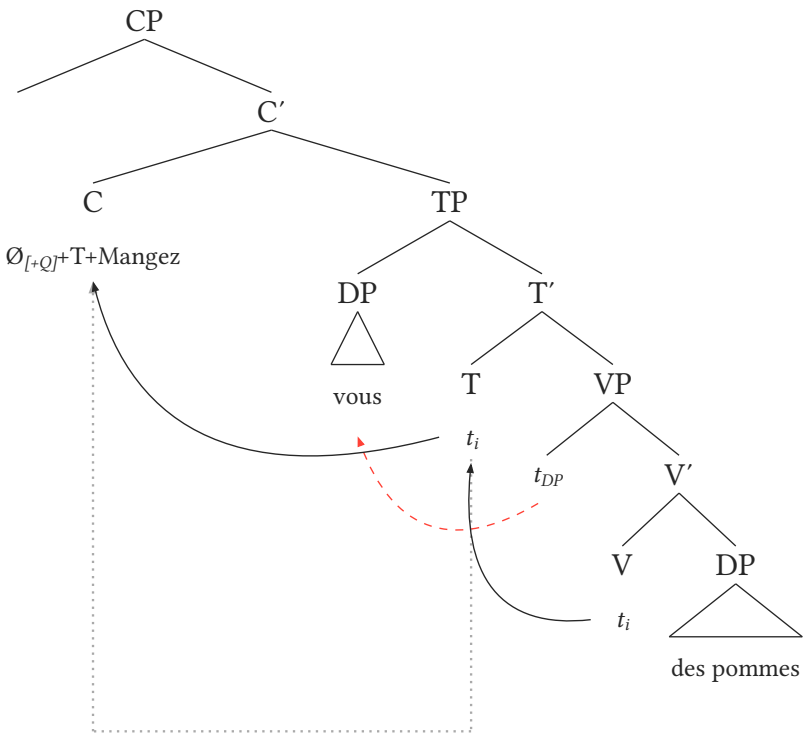
You will notice that some labels in Code 7 have a numeric suffix: `trace3-198`. That suffix is optional: it lets the user define exactly *where* the arrow should depart from (or arrive at, when it's used in `to`). Imagine an invisible ellipse surrounding each node's text. The numeric suffix represents a position on that ellipse in degrees, following standard trigonometric convention: 0° is east (right), 90° is north (up), 180° is west (left), and 270° is south (down). For example, `trace3-198` means "start the arrow from the 198° position around the `trace3` node" — slightly south of due west. Without a degree suffix, arrows use a default exit point in the tree's growth direction (typically below the text for downward trees), as can be seen in the gray dotted arrow once again — see first line inside `arrows` in Code 7. This gives the user a higher level of control overall.

The parameters `shift` and `bend` are easy to interpret, and allow the user to adjust the curve of each arrow independently. Both parameters have default values, which are implemented when `curved: true` and

no values are provided for `bend` or `shift`. The advantage of not specifying `curved`, which is a global parameter within the function, is that both rectangular and curved arrows can coexist in the same tree.

```
#tree(
    "[ CP [] [ C' [ C Ø_{[+Q]}+T+Mangez ]
          [ TP [ DP vous ] [ T' [ T t_i ]
            [ VP [ t_{DP} ] [ V' [V t_i ] [DP des pommes] ] ] ] ] ] ]",
    arrows: (
      (from: "trace1", to: "C1", dash: "dotted", color: gray, line-width: 1.70),
      (from: "trace3-198", to: "T1", dash: "solid", bend: 0.2, shift: -1),
      (from: "trace2-255", to: "DP1", dash: "dashed", bend: 1, shift: -0.5, color: red),
      (from: "trace1-556", to: "C1", dash: "solid", bend: 1, shift: -1.5),
    ),
  )
```

Code 7: Code for Tree 5



Tree 5: Multiple arrows, adapted from Carnie (2013)

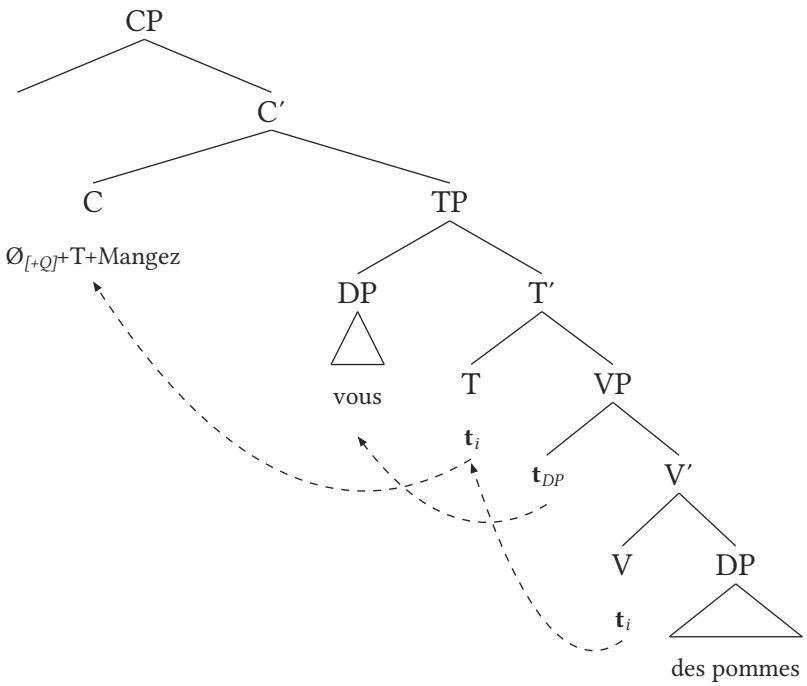
One final option regarding the specific location of arrows involves two other suffixes for labels. Traces are automatically labelled as `trace`, which means `trace3` in Tree 5 will target t_i, not the V node above it, which would be targeted with `v1`. By default, when we target a node `x`, the arrow targets its content, not the node itself. For phrase movements, we'd use a suffix: `xp-up`. The function assumes `-down` automatically, so `xp1` is the same as `xp1-down`. On top of that, we can append degrees to customize where an arrow hits, so `xp1-down-175` would mean "target the *content* of the first XP in the tree, specifically at 175°."

In summary, arrows have a lot of defaults, so you often end up with minimal code. However, they also allow for a relatively high degree of customization through the use of additional parameters. For reference, Tree 6 uses

all the default values for arrows, thus simplifying the code needed (Code 8) — `curved` is set to `true` here because multiple movements look better with curved arrows.

```
#tree(
  "[ CP [] [ C' [ C Ø_{[+Q]}+T+Mangez ] [ TP [ DP vous ] [ T' [ T *t*_i ] [ VP
[ *t*_{DP} ]
   [ V' [V *t*_i ] [DP des pommes] ] ]  ] ] ] ]",
  arrows: (
    (from: "trace3", to: "T1"),
    (from: "trace2", to: "DP1"),
    (from: "trace1", to: "C1"),
  ),
  curved: true,
)
```

Code 8: Code for Tree 6



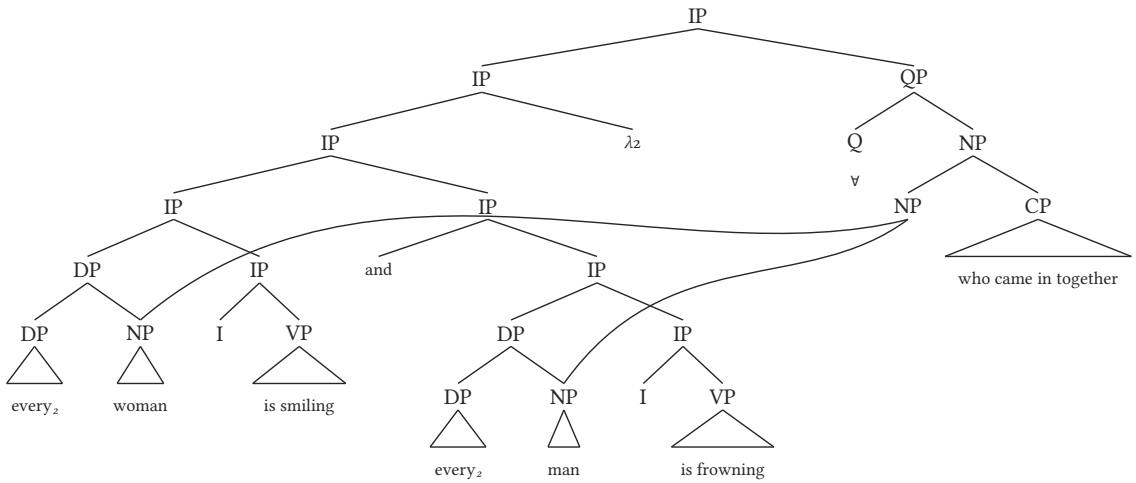
Tree 6: Multiple arrows, adapted from Carnie (2013)

4. Multidominance

Another important parameter in the function is `dominance`, which allows for more complex trees where multiple nodes are connected with curved branches. The example in Tree 7 comes from Fox & Johnson (2016, p. 7).

```
#tree(
  "[IP [IP [IP [IP [DP [DP every_2] [NP woman] ] [IP [I] [VP is smiling] ] ] [IP [and] [IP
[DP [DP every_2] [NP man] ] [IP [I] [VP is frowning] ] ] ] [\\lambda2] ] [QP [Q \
\forall] [NP [NP] [CP who came in together] ] ] ]",
  dominance: (
    (from: "NP4", to: "NP1", ctrl: (-4, 4.2)),
    (from: "NP4", to: "NP2", ctrl: (-3, 3.5)),
  ),
  scale: 0.7, // easily scale down tree to fit the document
  line-width: 1.5, // useful when tree is scaled down
  spread-local: ( // adjusts width locally, based on labels
    ("IP3", 0.6), // adjust the spacing for the third IP from the top (sister to lambda)
  ),
)
```

Code 9: Code for Tree 7



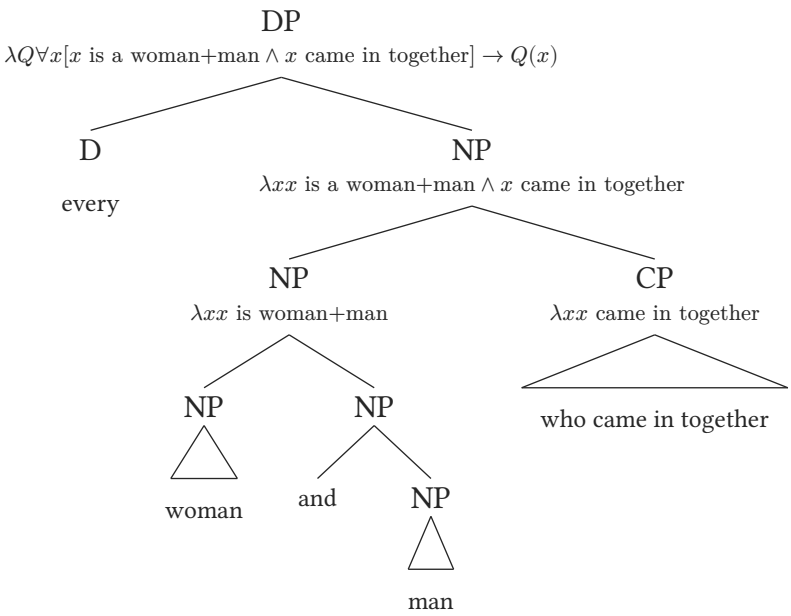
Tree 7: Multidominance

5. Semantics

A parameter is dedicated to additional content between a given node and its branches: `annotation`. This allows the user to add semantic derivations, for example. The parameter in question, shown in Code 10, uses **content blocks**, which means you can mix strings and math mode together with ease (unlike the `input` parameter for tree specification in the function). Math mode in Typst is more minimalistic than math mode in LaTeX (e.g., no backslashes are needed). Finally, `annotation-size` allows the user to set the font size for the annotations. The example in Tree 8 also comes from Fox & Johnson (2016, p. 7).

```
#tree(
  "[DP [D every] [NP [NP [NP woman] [NP [and] [NP man] ] ] [CP who came in together] ] ]",
  annotation: (
    ("DP1", [$lambda Q forall x [x "is a woman+man" and x "came in together"] arrow Q(x)$]),
    ("NP1", [$lambda x x "is a woman+man" and x "came in together"$]),
    ("NP2", [$lambda x x "is woman+man"$]),
    ("CP1", [$lambda x x "came in together"$]),
  ),
)
```

Code 10: Code for Tree 8

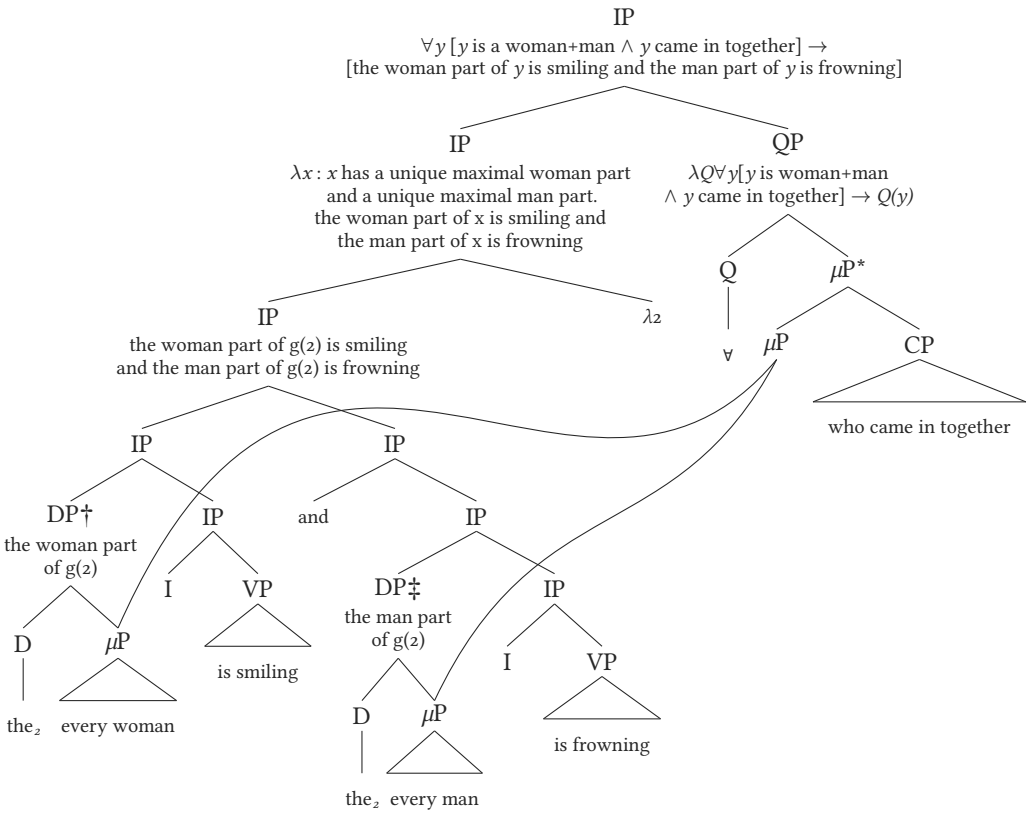


Tree 8: Semantic annotation with `annotation`

A final example includes longer annotation, from Fox & Johnson (2016, p. 13). Here, annotation involves multiple lines, so line breaks are manually added with `\`. You will notice that the code for the tree itself is relatively concise, but each annotation has its own entry, so Code 11 becomes longer. As you can see, any symbol can be directly added to a tree (e.g., ‡) — the symbol is then inherited by default by the label created for a given node, as can be seen in the annotation under DP‡ in Code 11.

```
  #tree(
  "[IP [IP [IP [IP [DP† [D the_2] [\\muP every woman] ] [IP [I] [VP is smiling] ] ]
      [IP [and] [IP [DP‡ [D the_2] [\\muP every man] ] [IP [I] [VP is frowning] ] ] ] ]
      [\\lambda2] ] [QP [Q \\forall ] [\\muP\\* [\\muP] [CP who came in together] ] ] ]",
  annotation: (
    (
      "IP1",
      [$forall$_y_ [_y_ is a woman+man $and$ _y_ came in together] $arrow$ \
        [the woman part of _y_ is smiling and the man part of _y_ is frowning]],
    ),
    (
      "IP2",
      [$lambda$_x_ : _x_ has a unique maximal woman part \
        and a unique maximal man part. \ the woman part of x is smiling and \
        the man part of x is frowning],
    ),
    (
      "QP1",
      [$lambda$_Q_ $forall$_y_[_y_ is woman+man \
        $and$ _y_ came in together] $arrow$ _Q(y)_],
    ),
    (
      "IP3",
      [the woman part of g(2) is smiling \ and the man part of g(2) is frowning],
    ),
    (
      "DP†1",
      [the woman part \ of g(2)],
    ),
    (
      "DP‡1",
      [the man part \ of g(2)],
    ),
  ),
  dominance: (
    (from: "muP4", to: "muP1", ctrl: (-6.1, 8.5)),
    (from: "muP4", to: "muP2", ctrl: (-6, 5)),
  ),
  scale: 0.8,
  spread: 0.8,
  terminal-branch: true,
)
```

Code 11: Code for Tree 9



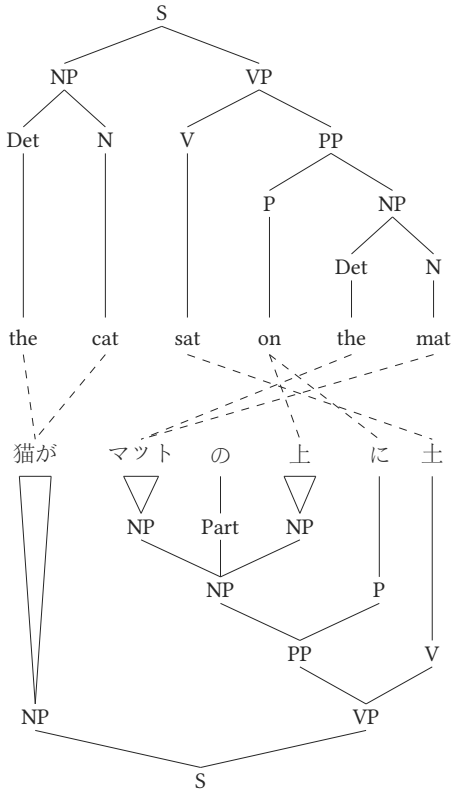
Tree 9: Complex tree with multi-line annotation

6. Bilingual trees

In one of the examples found in David Chiang's tutorial on `tikz-qtree` (Chiang, 2009), English and Japanese trees are compared in a single figure. This is accomplished here with the function `#garden()`, which allows for multiple trees to be built on top of each other. Because two is probably the ideal number of trees for this use case, that is the scenario illustrated in Tree 10. Here, again, the tree number is referenced by a suffix. For example, if we want to link the second N in tree 1 to the third NP in tree 2, we add `("n2-1", "np3-2")`. This is relatively intuitive, and automatic labels allow for Code 12 to be considerably more minimal than what we would need in LaTeX. Finally, to have a "mirror image" of two trees, one of the trees must be upside-down. This is why the `direction` parameter exists, discussed further in Section 10.4. We simply specify `direction: "up"` to flip a tree vertically, as shown in Tree 10.

```
#garden(
  ( // tree 1
    input: "[S [NP [Det the] [N cat]] [VP [V sat] [PP [P on] [NP [Det the] [N mat]]]]]",
    spread: 1.55, // This ensures that the trees' width is virtually the same
    content-size: 1
  ),
  ( // tree 2
  input: "[S [NP 猫が] [VP [PP [NP [NP マット] [Part の] [NP 上] ] [P に]] [V 土]]]",
  direction: "up",
  content-size: 1
  ),
    // Inter-tree lines (dashed)
  equivalence: (
    ("Det1-1", "NP1-2"), ("P1-1", "P1-2"), ("P1-1", "NP4-2"),
    ("N1-1", "NP1-2"), ("N2-1", "NP3-2"), ("Det2-1", "NP3-2"),
    ("V1-1", "V1-2"),
  ),
  gap: 2.5,
  scale: 0.7
)
```

Code 12: Code for Tree 10



Tree 10: Bilingual trees — example from Chiang's tutorial on `tikz-qtree`

```
\begin{tikzpicture}
   \begin{scope}[frontier/.style={distance from root=150pt}]
   \Tree [.S [.NP [.Det \node(e1){the}; ]
             [.N \node(e2){cat}; ] ]
           [.VP [.V \node(e3){sat}; ]
             [.PP [.P \node(e4){on}; ]
               [.NP [.Det \node(e5){the}; ]
                 [.N \node(e6){mat}; ] ] ] ] ]
   \end{scope}
   \begin{scope}[xshift=9pt,yshift=-5in,grow'=up,
             frontier/.style={distance from root=150pt}]
   \tikzset{every leaf node/.style={font=\ja}}
   \Tree [.S [.NP \node(j1){猫が}; ]
           [.VP [.PP [.NP [.NP \node(j2){マット}; ]
                     [.Part \node(j3){の}; ]
                     [.NP \node(j4){上}; ] ]
                 [.P \node(j5){に}; ] ]
             [.V \node(j6){土}; ] ] ]
   \end{scope}
   \begin{scope}[dashed]
      \draw (e1)--(j1);
      \draw (e2)--(j1);
      \draw (e3)--(j6);
      \draw (e4)--(j4);
      \draw (e4)--(j5);
      \draw (e5)--(j2);
      \draw (e6)--(j2);
   \end{scope}
\end{tikzpicture}
```

Code 13: Equivalent code in LaTeX to produce Tree 10

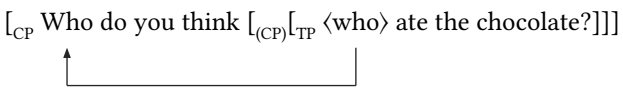
By default, stacked trees will use `bottom: true` to ensure that alignment is optimal — see Tree 17 in Section 10.1. Consequently, terminal branches are displayed to avoid empty space between nodes and their content.

7. In-line movement

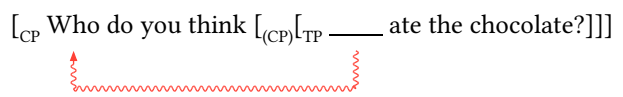
In-line representations also make use of automatic labelling. Every word we add to a sentence is its own label. As a result, we can easily draw an arrow from `who2` to `who1` in Example 1. Because the arrows extend below the text baseline, they may overlap with subsequent content. Setting `protect: true` reserves vertical space below the output to prevent this. By default, `protect` is `false`, which is the correct setting when `#move()` is placed inside a table cell (e.g., in numbered examples with `#eg()`), where extra height would misalign adjacent cells — see Section 8 below. Because these representations are almost always used in the context of a numbered example, `protect: false` is favoured.

The examples below demonstrate how to change line styles for arrows and how to delink a given arrow using its position. For example, if `delinks: (0,)`, a delink is added to the first movement arrow added to the code — see last two examples in Example 1.

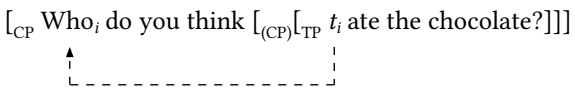
```
#move("[CP Who do you think [(CP)[TP<who>ate the chocolate?]]]",
  arrows: ((from: "who2", to: "who1"),),
  protect: true,)
```



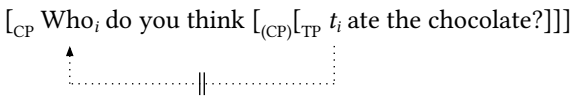

```
#move("[CP Who do you think [(CP)[TP __ ate the chocolate?]]]",
  arrows: ((from: "trace1", to: "who1", dash: "wavy", color: red),),
  protect: true,)
```



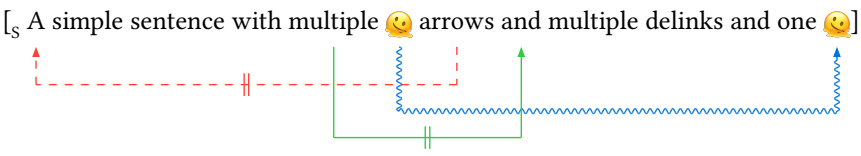

```
#move("[CP Who do you think [(CP)[TP t_i ate the chocolate?]]]",
  arrows: ((from: "trace1", to: "who1", dash: "dashed"),),
  protect: true,)
```




```
#move("[CP Who do you think [(CP)[TP t_i ate the chocolate?]]]",
  arrows: ((from: "trace1", to: "who1", dash: "dotted"),),
  delinks: (0,), // delink the first (0) arrow
  protect: true,)
```




```
#move("[S A simple sentence with multiple 😉 arrows and multiple delinks and one 😉]",
  arrows: (
    (from: "simple1", to: "show1", dash: "dotted", color: black),
    (from: "arrows1", to: "a1", dash: "dashed", color: red),
    (from: "😉1", to: "😉2", dash: "wavy", color: blue),
    (from: "multiple1", to: "and1", dash: "solid", color: green),
  ),
  delinks: (0, 2),
  protect: true,)
```



Example 1: In-line movement, traces, arrows and delinks (emojis also work with labels)

8. Numbered examples

The examples shown in Example 1 are usually placed in numbered examples. This can be accomplished with the function `#eg()`, which is the same implementation found in **phonokit**, to keep both packages consistent.

> **Important.** To use this function, add this line after importing the package: `#show: eg-rules`

The simplest use of `#eg()` is to wrap content directly. The example number is generated automatically:

```
#eg[#move(
  "[CP Who do you think [(CP)[TP<who>ate the chocolate?]]]",
  arrows: ((from: "who2", to: "who1"),),
)] <eg-single>
```

Code 14: Single-item numbered example

(1) [$_{CP}$ Who do you think [$_{(CP)}$[$_{TP}$ ⟨who⟩ ate the chocolate?]]]

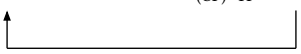
For sub-examples, use the list syntax (`-`). Each item is automatically lettered (`a.`, `b.`, etc.). The `labels` argument allows individual sub-examples to be referenced, as in (2a) and (2b). The example as a whole can also be referenced, as in (2).

```
#eg(labels: (<s-plain>, <s-move>))[
  - Who do you think saw Mary?
  - #move(
      "[CP Who do you think [(CP)[TP<who>saw Mary]]]",
      arrows: ((from: "who2", to: "who1", dash: "solid", color: black),),
    )
] <eg-wh>
```

Code 15: Sub-examples with labels

(2) a. Who do you think saw Mary?

b. [$_{CP}$ Who do you think [$_{(CP)}$[$_{TP}$ ⟨who⟩ saw Mary]]]



Without `labels`, sub-examples are simply not "referenceable" individually — only the example as a whole can be labelled and referenced. The `title` argument places a title on the same line as the example number, with the content below. Finally, the `caption` argument is only used if you plan to have a table of contents exclusively for examples, which covers both `#eg()` and `#gloss()` (discussed next). In (3), a `caption` has been added, but nothing is printed. In Section 9, an outline is produced where this caption will be useful.

```
#eg(title: [Wh-movement in English], caption: [Wh-movement in English])[
  - Who do you think saw Mary?
  - #move(
      "[CP Who do you think [(CP)[TP<who>saw Mary]]]",
      arrows: ((from: "who2", to: "who1"),),
    )
] <eg-title>
```

Code 16: Example with `title` to generate (3)

(3) Wh-movement in English

a. Who do you think saw Mary?

b. $[_{CP}$ Who do you think $[_{(CP)}[_{TP}$ ⟨who⟩ saw Mary]]]

A note on figure spacing. This applies to **phono*kit*** as well as to **syn*kit***: If your document uses a custom `#show figure` rule to add vertical spacing around figures, you may notice misalignment in sub-example labels. This happens because `#subex-label()` uses figures internally, and added spacing will disrupt table cell alignment. To fix this, exclude `linguistic-subexample` from your spacing rule. For example, a document that adds `1em` vertical spacing before and after each figure should have something along these lines:

```
#show figure: it => {
  if it.kind == "linguistic-subexample" {
    it
  } else {
    v(1em)
    it
    v(1em)
  }
}
```

Code 17: Excluding linguistic examples from custom figure spacing specifications

9. Glosses

The `#gloss()` function is meant to be a lightweight companion to `#eg()`, especially when examples need to share numbering across **syn*kit*** and **phono*kit***. For a more complete glossing solution, I strongly recommend using the `eggs` package (Shaldov, 2025), which is a dedicated package.

For numbered examples involving glosses, where alignment is essential, you can use the `#gloss()` function. Its implementation is simple (similar to `#eg()` discussed above): it uses regular expressions to look for spaces, which trigger a split that guarantees alignment across words. Its syntax is also minimal, since it's simply a list. Curly braces can be used to trigger small caps. Both functions (`#gloss()` and `#eg()`) share the same numbering, as expected, since both are numbered examples. See Code 18, which generates (4) (note that a `caption` is added here, which will be relevant later in this section).

```
#gloss(spacing: 1.2em, caption: [An example from Portuguese])[
  - eu gosto de maçã
  - I like-{1sg.prs} of apple
  - 'I like apples.'
] <eg-gloss-1>
```

Code 18: Code to generate (4)

(4) eu gosto de maçã
I like-1SG.PRS of apple
'I like apples.'

Labels such as `<eg-gloss-1>` seen in Code 18 are global, but you can also add labels for subglosses (exactly like `#eg()`) with the `labels` argument. Likewise, a `per` argument exists in case you want to have multiple glosses in the same example with a custom number of lines for whatever reason. The reason for `per` is flexibility: if you wanted to have glosses with 5 lines each (line 5 being the translation), you would simply define `per: 5`. Code 19 shows how (5) is generated. Given that `labels` have been added in Code 19, we can now refer to (5a) and (5b).

```
#gloss(per: 3, labels: (<eg-sub-pt1>, <eg-sub-pt2>))[
  - eu gosto de maçã
  - I like.{1sg.prs} of apple
  - 'I like apples.'
  - elle aime les pommes
  - she like.{3sg.prs} {def.pl} apple.{pl}
  - 'She likes apples.'
] <eg-sub-pt>
```

Code 19: Code to generate (5)

(5) a. eu gosto de maçã
I like.1SG.PRS of apple
'I like apples.'

b. elle aime les pommes
she like.3SG.PRS DEF.PL apple.PL
'She likes apples.'

Let's explore some additional cases. In (6), we see an example from Inuktitut (O'Grady & Archibald, 2017, p. 275). This is a situation where we may want to have four lines for a gloss, where a given line simply shows an orthographic form that should not be parsed by the `#gloss()` function. That's why the `escape` argument exists. In (6), the first line (number `0`) is "free" from the parsing used elsewhere in the example.

```
#gloss(per: 4, escape: (0,), caption: [An example from Inuktitut])[
  - Qasuiirsarvigssarsingitluinarnarpuq
  - Qasu -iir -sar -vig -ssar -si -ngit-luinar -nar -puq
  - tired not cause-to-be place-for suitable find not-completely someone 3.{sg}
  - 'Someone did not find a completely suitable resting place.'
] <gloss-inuktitut>
```

Code 20: Code to generate (6)

(6) Qasuiirsarvigssarsingitluinarnarpuq
Qasu -iir -sar -vig -ssar -si -ngit-luinar -nar -puq
tired not cause-to-be place-for suitable find not-completely someone 3.SG
'Someone did not find a completely suitable resting place.'

Here's another case, this time from Turkish (O'Grady & Archibald, 2017), where we have IPA symbols[1] and only two lines (`per: 2`). This is shown in (7).

```
#gloss(per: 2)[
  - [kœj]
  - 'village'
] <gloss-turkish>
```

Code 21: Code to generate (7)

(7) [kœj]
'village'

Finally, a case with a title, also from Turkish (O'Grady & Archibald, 2017), is shown in (8).

```
#gloss(title: [SOV (Turkish)], caption: [Another example])[
  - Hasan œkyz-y al-dɨ
  - Hasan ox-{acc} buy-{pst}
  - 'Hasan bought the ox.'
] <gloss-turkish2>
```

Code 22: Code to generate (8)

(8) SOV (Turkish)
Hasan œkyz-y al-dɨ
Hasan ox-ACC buy-PST
'Hasan bought the ox.'

[1] Because Typst has native UTF-8 support, you can simply add whatever symbol you want.

We can now generate a simple table of contents that targets only the examples for which we set a `caption` above. These include both `#eg()` and `#gloss()`. If you don't add a `caption` for an example, it will not appear in the table of contents generated below (see Code 23).

```
#outline(
  title: [List of Examples],
  target: figure.where(kind: "linguistic-example"),
)
```

Code 23: Code to create an outline for examples

List of Examples

If you also use **phonokit** in your document, the outline above will integrate phonology and syntax examples seamlessly, even though both packages use two different functions, namely `#ex()` and `#eg()`.

10. Extras

This section introduces some important parameters to further adjust syntax trees.

10.1. Spacing

Table 1 lists key parameters to understand how spacing works in `#tree()`.

Parameter	Purpose
`spread`	Global spacing between sister nodes (horizontal)
`spread-local`	Local spacing between sister nodes (based on labels)
`drop`	Global spacing between levels (vertical)
`drop-local`	Local spacing between levels (based on labels)

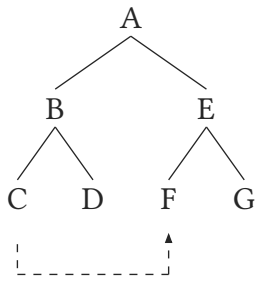
Table 1: Spacing parameters in `#tree()`

To see these parameters in action, let's examine Tree 11 to Tree 16. These are all based on the base code shown in Code 24. Tree 11 shows what a typical tree looks like without any custom specification for the parameters in Table 1. An arrow is added to all trees to show the user how they adjust automatically as we adjust any spacing parameter (since they're based on labels and not on absolute coordinates).

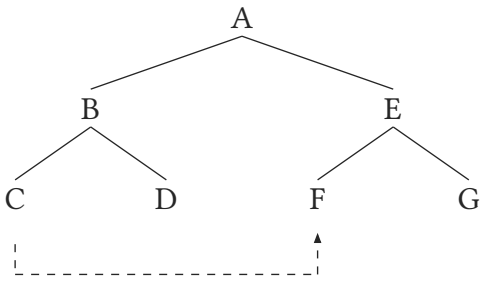
If we want the tree as a whole to have more horizontal separation, `spread` can be used. Tree 12, for example, increases the spread by 100% (from `1` to `2`). But we often want to adjust only one sister-to-sister interval; that's why `spread-local` exists, as shown in Tree 13, which reduces the gap after C (that is, the C-D spacing). The parameter `drop` works on the same direct scale for vertical spacing, as can be seen in Tree 14 to Tree 16.

```
#tree("[ A [ B [C] [D] ] [ E [F] [G] ] ]", arrows: ("C1", "F1"), .. .. ..)
```

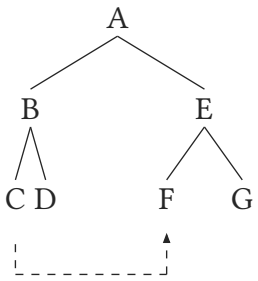
Code 24: Code for Tree 11 through Tree 16



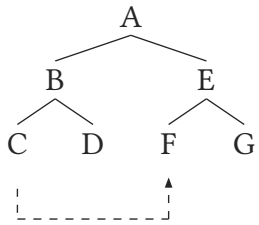
Tree 11: Default



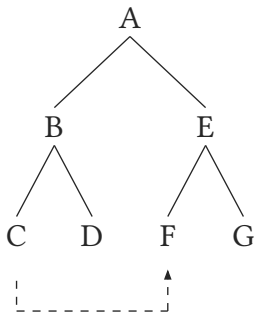
Tree 12: `spread: 2`



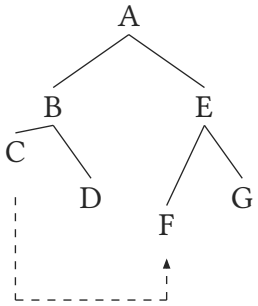
Tree 13: `spread-local` (C-D)



Tree 14: `drop: 0.7`



Tree 15: `drop: 1.2`

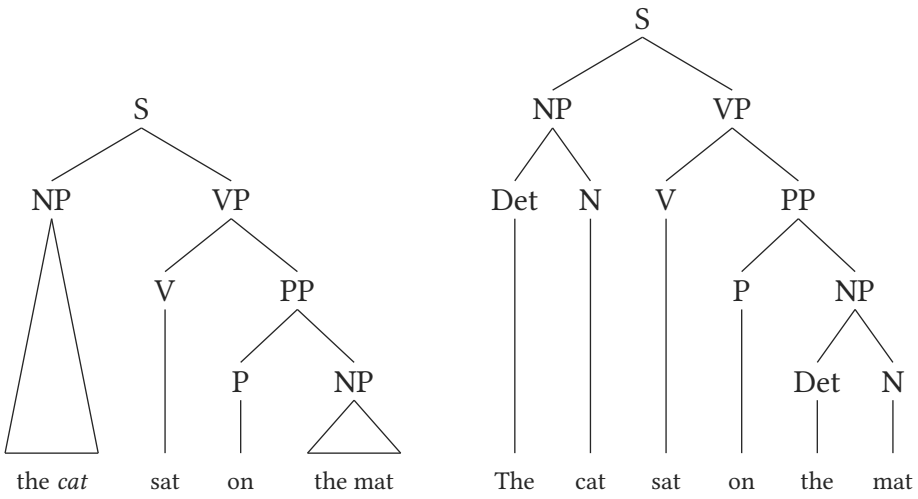


Tree 16: `drop-local` (C-F)

Another possible adjustment to spacing involves the vertical alignment of node contents, i.e., the spacing between a node and its content. The parameter `bottom: true` will pull the tree's content down and bottom-align it. This will trigger terminal branches by default, as already mentioned.

```
#tree(
  "[S [NP the *cat*] [VP[V sat][PP[P on] [NP the mat]]]]",
  bottom: true,
)
#tree(
  "[S [NP [Det The] [N cat] ] [VP[V sat][PP[P on] [NP [Det the] [N mat]]]]]",
  bottom: true,
)
```

Code 25: Code for Tree 17



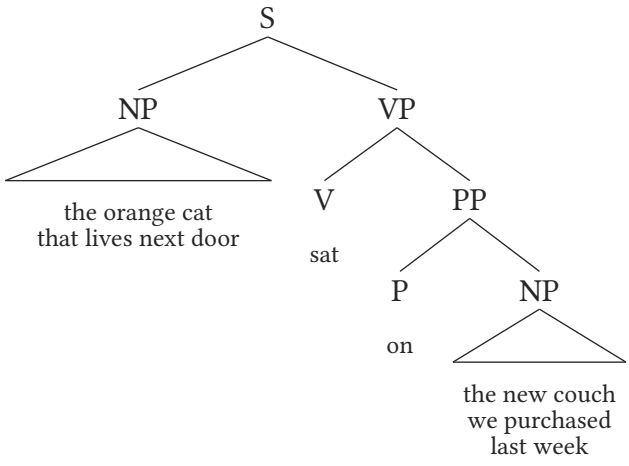
Tree 17: Bottom alignment — looks better without triangles

10.2. Line breaks

Breaking lines is easy with `\\n`. This avoids having phrases that are too wide.

```
#tree(
    "[S [NP the orange cat \\n that lives next door]
      [VP[V sat][PP[P on]
      [NP the new couch \\n we purchased \\n last week]]]]",
  )
```

Code 26: Code for Tree 18



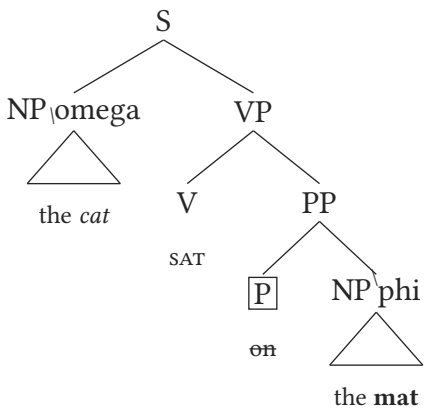
Tree 18: Breaking lines within phrases

10.3. Formatting

Let's now explore some important consequences of using a string as our main input. What if we want to use bold, italics, small caps, Greek letters, etc.? If you have used Typst before, you know that you can simply add whatever symbol you need directly into the document (UTF-8 is natively accepted). But `#tree()` also has some presets to make this easier. Tree 19 shows how to adjust the formatting of strings in the tree. Greek letters are interpretable as long as you use `\\`. For italics, you use `*cat*`; for bold, `**mat**`, for small caps, `@sat@`. You can also strike through text with `~` (see Tree 19). Finally, subscripts and superscripts can be easily added with `_` and `^`, respectively. An additional parameter, `highlight`, exists for drawing a rectangle around any given node (equivalent to `\node[draw]` in `tikz-qtree`) — we refer to `P` as `"P1"` in Code 27 ("first [and only] P from the top of the tree").

```
#tree("[S [NP_\\omega the *cat*] [VP[V @sat@][PP[P ~on~] [NP^\\phi the **mat**]]]]",
            highlight: ("P1",))
```

Code 27: Code for Tree 19



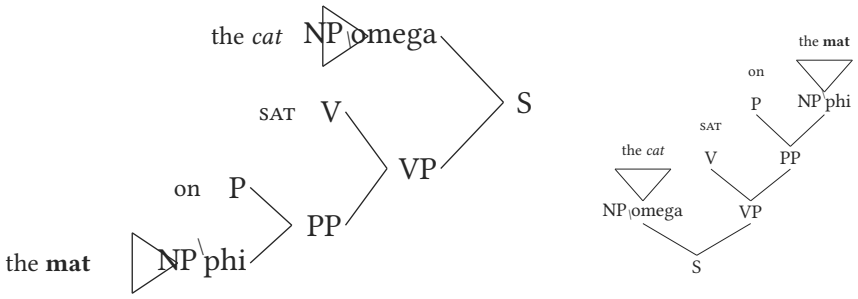
Tree 19: The basics: highlight, bold, italics, small caps, super- and subscripts

10.4. Direction

Trees can also change direction with the `direction` parameter. By default, `direction: "down"`, but you can set it to `"up"`, `"left"`, or `"right"`, as shown in Code 28. In addition, you can use the `scale` parameter to increase or decrease the size of your tree. In Tree 20, the tree on the left uses `scale: 1` (default), while the tree on the right uses `scale: 0.6` (60% of its original size).

```
#tree("[S [NP_\\omega the *cat*] [VP[V @sat@][PP[P on] [NP^\\phi the **mat**]]]]",
            direction: "left")
```

Code 28: Code for Tree 20



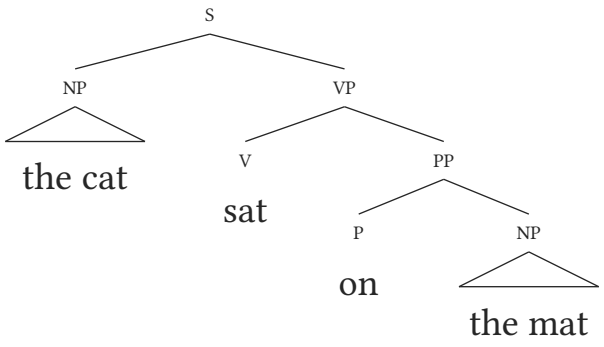
Tree 20: Changing tree direction and size

10.5. Sizing

Sizing and spacing are two key parameters in any tree. By default, the content of each node has 80% of the size of the node itself, but this can be changed with the `content-size` parameter (defaults to `0.8`). Tree 21 uses `content-size: 1.2` to make the content larger than the nodes. In Code 29, you will also find two other parameters: `drop` and `spread`. These help you set vertical and horizontal spacing for the tree. If you wish to change the size of nodes, `node-size` is also available.

```
#tree(
  "[S [NP the cat] [VP[V sat][PP[P on] [NP the mat]]]]",
  content-size: 1.2,
  node-size: 0.6,
  drop: 0.8,
  spread: 1.5,
)
```

Code 29: Code for Tree 21



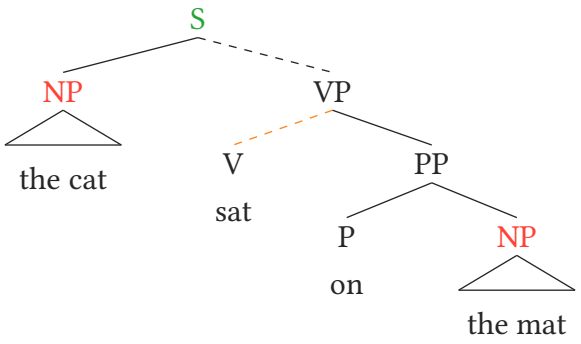
Tree 21: Size and spacing: larger content, smaller nodes

10.6. Branch styling

Different parameters exist for styling a tree, most notably `color` and `dash-branches`. Tree 22 uses both parameters. The parameter `color` is flexible and can refer to both branches or text, depending on the parameters passed to it. For example, `color: ("S1", green)` will target a node (S), while `color: ("VP1", "V1")` will target *the relationship* between nodes, i.e., a branch.

```
#tree(
  "[S [NP the cat] [VP[V sat][PP[P on] [NP the mat]]]]",
  content-size: 1,
  drop: 0.8,
  spread: 1.5,
  dash-branches: (
    ("VP1", "V1"),
    ("S1", "VP1"),
  ),
  color: (
    ("S1", green.darken(20%)),
    ("NP1", red),
    ("NP2", red),
    ("P1down", blue),
    ("VP1", "V1", orange),
  ),
)
```

Code 30: Code for Tree 22



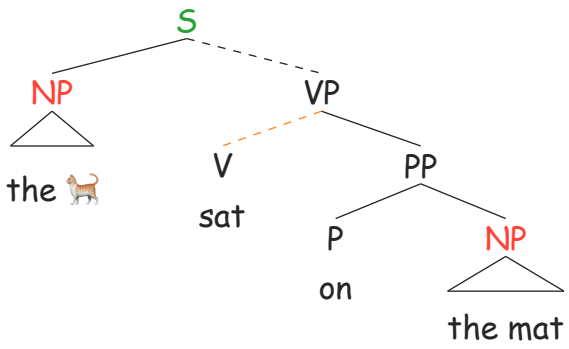
Tree 22: A colourful tree

10.7. Font

While **synkit** inherits the document font by default, you can always choose whatever font you prefer for each function. Tree 23 illustrates this by setting `font: "Comic Sans MS"`.

```
#tree(
  "[S [NP the 🐈] [VP[V sat][PP[P on] [NP the mat]]]]",
  content-size: 1,
  drop: 0.8,
  spread: 1.5,
  dash-branches: (
    ("VP1", "V1"),
    ("S1", "VP1"),
  ),
  color: (
    ("S1", green.darken(20%)),
    ("NP1", red),
    ("NP2", red),
    ("P1down", blue),
    ("VP1", "V1", orange),
  ),
  font: "Comic Sans MS",
)
```

Code 31: Code for Tree 23



Tree 23: Changing the font locally (emojis natively supported)

11. Future work and acknowledgments

The package is in its infancy, so there will be bugs and limitations. That is why comments and suggestions are especially welcome, as they will speed up the process of improving precision and coverage. Simply open an issue in the package's repo. I am not a syntactician, so I've benefitted greatly from the help of my colleague Vincent Rouillard.

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Appendix

A. Argument reference

Table 2 lists all available parameters in `#tree()`.

Argument	Type	Default	Description
`arrows`	array	`()`	Cross-node arrows. Each entry is `(from, to)` or a dict with keys `from`, `to`, `color`, `bend`, `shift`, `dash`, `line-width`
`scale`	number	`1.0`	Uniform scale factor for the whole tree
`triangle`	array	`()`	Anchor names whose branch renders as a triangle for elided structure
`content-size`	number	`0.8`	Size multiplier for leaf content text
`node-size`	number	`1.0`	Size multiplier for node label text
`curved`	bool	`false`	Draw arrows as Bézier curves instead of rectangular paths
`direction`	string	`"down"`	Growth direction: `"down"`, `"up"`, `"right"`, or `"left"`
`highlight`	array	`()`	Anchor names to draw a box around. Use `"DP1-down"` to box the leaf content
`bottom`	bool	`false`	Align all leaves at the lowest level regardless of depth
`terminal-branch`	bool	`false`	Draw branches from non-terminal nodes to terminal (leaf) nodes
`dash-branches`	array	`()`	Array of `(parent, child)` anchor pairs whose branch is dashed
`delinks`	array	`()`	Anchor names where a delink mark (two bars) is drawn on the arrow shaft
`index`	array	`()`	Coreference subscripts. Array of single-key dicts, e.g. `(("CP1": "i"),)`
`append`	array	`()`	Extra subscript text appended after a node label, e.g. `(("VP1", "0"),)`
`drop`	number	`1.0`	Multiplier for vertical distance between levels
`drop-local`	array	`()`	Per-level or per-node branch length multipliers, e.g. `(1, 1.5)` or `("IP2", 0.5)`
`spread`	number	`1.0`	Horizontal width per leaf in canvas units
`spread-local`	array	`()`	Per-level or per-node horizontal gap multipliers between sisters
`dominance`	array	`()`	Long-distance dominance lines. Entries: `("from", "to")` or dict with `ctrl`, `color`, `dash`
`color`	array	`()`	Colorize nodes, leaves, or branches. E.g. `("NP1", red)` or `("VP1", "V1", yellow)`
`annotation`	array	`()`	Semantic annotations between node label and branches, e.g. `(("DP1", $lambda$),)`
`annotation-size`	number	`0.70`	Size multiplier for annotation text
`annotation-leading`	length or `auto`	`auto`	Line spacing for multi-line annotations
`numbers`	array	`()`	Circled numbers placed to the left of node labels, e.g. `(("DP1", 2),)`
`numbers-size`	number	`0.85`	Size multiplier for circled numbers relative to node labels
`line-width`	number	`1.0`	Stroke width multiplier for all tree lines
`font`	string	`none`	Font family used in the function (local scope)

Table 2: All arguments in `#tree()`

B. How can I use Typst offline?

This vignette assumes that you know about Typst, but you may not be very familiar with it. That's why this section exists. For example, while the online app at Typst.app is very useful and practical, most of us prefer to work offline. **How can you use Typst offline then?**

One of the best IDE options out there is to use VS Code with the extension Tinymist (Dreamin & Varner, 2024) — the extension is therefore available for Positron, which is the successor to RStudio. Tinymist is also available as a plugin for NeoVim users. All these options work extremely well because Tinymist is great, and I haven't had any issues thus far: compilation is instantaneous, and `bib` files also work flawlessly.[2]

C. How do packages work in Typst?

If you've used R, Python, LaTeX, etc., you are used to installing packages and then importing them. This vignette has imported **synkit**, of course, which in turn imports CeTZ (Fejfar & Funke, 2024) as a dependency. As you start using Typst, you will notice that it works a bit differently, and this may not be self-evident at first. As seen in Section 1.1, there are basically two ways to load and use a package, both of which require the function `#import` inside your `typ` document — notice that you don't install a package *per se*. The traditional way is to import a package from the official Typst collection/repository, which means adding `#import "@preview/synkit:0.0.2": *` to your `typ` document if you plan on using **synkit** (assuming version `0.0.2`). The `@preview` bit indicates that the package comes from Typst's official repository. This is what you should do most of the time. Typst packages are cached once you compile a document with a given package.

Another option is to fork, clone or download a package from GitHub and import its `lib.typ` file instead: `#import "PACKAGE_DIRECTORY/lib.typ": *`. There's only one caveat: Typst restricts imports to files within the compilation root and its subdirectories (i.e., you can't load `lib.typ` if the package is in a parent directory or elsewhere in your system). Thus, you may need to use symlinks (this is the same strategy applied to `bib` files if you don't want to have a local copy of your references).

```
# From your working directory:
ln -s PATH_TO_PACKAGE_DIRECTORY package_name
```

Code 32: Creating a symlink to local package

Finally, Typst's repository contains sub-directories to keep track of each version of a given package. When a package is updated, nothing happens to the existing version of the package. Instead, a new directory is added with the updated version. That's why you specify the *version* of a package upon importing: `#import "@preview/synkit:0.0.2": *`. If you go to Typst's repository on GitHub, you will see that the repository for **synkit** has one sub-directory for each version of the package. This means that you always know which version of a package you are using. And because previous versions are not removed, backwards compatibility is not an issue. If you are used to LaTeX and you have the habit of frequently updating your packages, you will appreciate this, as there's no risk of recompiling your document and running into errors because one of the packages you use has been updated with breaking changes.

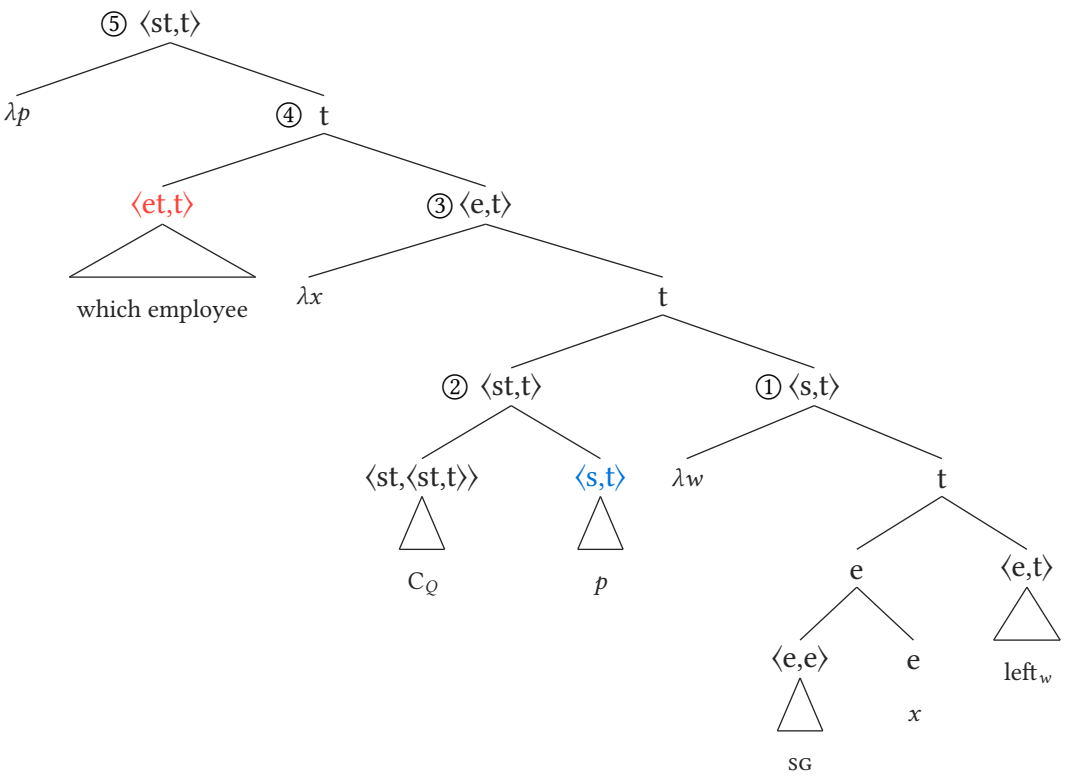
[2] Provided that they are clean and do not have any problems regarding fields, repeated entries, etc.

D. Additional trees

Example from Elliott et al. (2022, p. 10) showing the flexibility of labels and the `numbers` argument.

```
#tree(
  "[<st,t> [\\lambda*p*] [t [<et,t> which employee] [<e,t> [\\lambda*x*]
      [t [<st,t> [<st,<st,t>> C_Q] [<s,t> *p* ] ] [<s,t> [\\lambda*w*]
          [t [e [<e,e> @sg@ ] [e *x*] ] [<e,t> left_w] ] ] ] ] ] ]",
    color: (
      ("ett1", red),
      ("st1", blue),
    ),
    numbers: ( // Label stripping ignores <, >, and , in node names
      ("stt1", 5),
      ("t1", 4),
      ("et1", 3),
      ("stt2", 2),
      ("st2", 1),
    ),
    numbers-size: 1.2,
)
```

Code 33: Code for Tree 24



Tree 24: Numbers and flexible labels (example with colors)